

## STATESBORO, GA, USA

WMO: 747805

Lat: 32.483N

Lon: 81.737W

Elev: 187

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63818

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.2       | 30.0 | 6.5                           | 7.7 | 40.7 | 13.5 | 11.0 | 42.3 | 20.8                  | 53.4 | 18.8 | 52.7 | 2.3                   | 320  | 0.354 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.0                   | 97.4            | 78.4 | 95.4 | 77.6 | 93.2 | 77.6 | 85.2                | 91.7 | 83.6 | 89.6 | 81.8 | 85.7 | 5.8                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 83.7                            | 177.5 | 90.2 | 81.9 | 167.1 | 86.7 | 81.0 | 162.1 | 85.1 | 49.6          | 90.9 | 47.7 | 89.7 | 45.6 | 87.4 | 91.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 17.1              | 14.4 | 11.9 | 21.5                       | 99.7 | 3.5                | 3.0 | 19.0                                               | 101.8 | 16.9       | 103.6 | 15.0       | 105.3 | 12.4       | 107.4 |
|                   |      |      | 19.8                       | 83.4 | 3.6                | 3.7 | 17.2                                               | 86.1  | 15.0       | 88.3  | 13.0       | 90.4  | 10.4       | 93.1  |
|                   |      |      | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 67.3        | 49.4  | 53.3  | 59.8  | 66.8  | 74.8  | 81.2  | 83.1  | 82.5  | 78.0  | 68.1  | 57.1  | 52.6  | (6)  |
|                                                                        | DBStd       | 13.87       | 9.61  | 8.73  | 8.49  | 6.83  | 5.70  | 3.87  | 3.25  | 3.44  | 4.36  | 7.18  | 8.03  | 8.98  | (7)  |
|                                                                        | HDD50       | 308         | 133   | 58    | 19    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 23    | 74    | (8)  |
|                                                                        | HDD65       | 1803        | 489   | 339   | 203   | 60    | 5     | 0     | 0     | 0     | 0     | 51    | 259   | 397   | (9)  |
|                                                                        | CDD50       | 6625        | 114   | 150   | 323   | 506   | 769   | 935   | 1027  | 1008  | 839   | 561   | 237   | 156   | (10) |
|                                                                        | CDD65       | 2644        | 5     | 11    | 41    | 115   | 309   | 485   | 562   | 543   | 389   | 147   | 23    | 14    | (11) |
|                                                                        | CDH74       | 26408       | 31    | 99    | 438   | 1207  | 3047  | 5075  | 6089  | 5592  | 3403  | 1185  | 173   | 69    | (12) |
|                                                                        | CDH80       | 11667       | 0     | 8     | 84    | 356   | 1312  | 2472  | 3004  | 2643  | 1433  | 328   | 21    | 6     | (13) |
| (14) Wind                                                              | WSAvg       | 4.1         | 4.9   | 5.0   | 5.1   | 5.1   | 4.4   | 3.8   | 3.4   | 3.0   | 3.2   | 3.4   | 3.7   | 4.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 47.5        | 4.4   | 3.3   | 4.0   | 2.6   | 2.7   | 5.8   | 5.3   | 5.9   | 3.8   | 3.6   | 2.7   | 3.0   | (15) |
|                                                                        | PrecMax     | 64.1        | 9.4   | 9.4   | 8.6   | 6.1   | 7.9   | 12.0  | 10.7  | 12.8  | 7.1   | 12.3  | 7.5   | 6.0   | (16) |
|                                                                        | PrecMin     | 29.8        | 0.8   | 0.4   | 0.3   | 0.5   | 0.2   | 1.9   | 2.5   | 2.3   | 0.3   | 0.1   | 0.6   | 0.5   | (17) |
|                                                                        | PrecStd     | 10.0        | 2.2   | 2.0   | 2.6   | 1.5   | 1.8   | 2.8   | 2.2   | 3.0   | 1.9   | 3.2   | 1.9   | 1.3   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 77.8  | 81.3  | 86.2  | 90.4  | 97.0  | 99.5  | 99.7  | 99.7  | 95.2  | 90.2  | 82.4  | 80.6  | (19) |
|                                                                        |             | MCWB        | 66.5  | 65.4  | 69.0  | 71.5  | 74.9  | 75.8  | 80.7  | 81.1  | 74.7  | 75.1  | 70.0  | 69.6  | (20) |
|                                                                        | 2%          | DB          | 73.1  | 77.3  | 82.0  | 87.5  | 93.0  | 97.1  | 97.2  | 96.9  | 92.6  | 86.7  | 79.2  | 75.3  | (21) |
|                                                                        |             | MCWB        | 64.3  | 63.3  | 65.9  | 69.6  | 73.4  | 76.5  | 79.9  | 79.3  | 75.3  | 72.9  | 68.3  | 68.6  | (22) |
|                                                                        | 5%          | DB          | 69.6  | 73.1  | 79.2  | 84.1  | 90.4  | 94.4  | 95.0  | 93.5  | 90.5  | 83.7  | 75.2  | 72.1  | (23) |
|                                                                        |             | MCWB        | 62.5  | 61.2  | 64.4  | 67.3  | 72.1  | 76.7  | 79.2  | 78.6  | 75.1  | 70.9  | 66.1  | 66.3  | (24) |
|                                                                        | 10%         | DB          | 65.5  | 69.6  | 75.2  | 81.4  | 87.8  | 91.2  | 92.8  | 91.3  | 88.3  | 81.1  | 72.2  | 68.1  | (25) |
|                                                                        |             | MCWB        | 60.6  | 59.8  | 62.0  | 65.6  | 71.4  | 76.3  | 79.3  | 77.7  | 74.9  | 70.0  | 64.0  | 63.5  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 70.1  | 69.4  | 73.5  | 76.4  | 81.6  | 85.0  | 87.0  | 88.9  | 84.0  | 81.5  | 75.5  | 74.5  | (27) |
|                                                                        |             | MCDB        | 73.4  | 74.3  | 79.9  | 84.2  | 88.9  | 91.8  | 94.9  | 96.0  | 89.2  | 84.9  | 80.0  | 76.4  | (28) |
|                                                                        | 2%          | WB          | 67.1  | 67.0  | 70.0  | 73.4  | 78.7  | 83.1  | 85.3  | 85.3  | 81.3  | 77.8  | 72.2  | 70.4  | (29) |
|                                                                        |             | MCDB        | 70.9  | 73.2  | 76.1  | 80.1  | 86.1  | 89.6  | 92.8  | 90.1  | 84.2  | 80.5  | 75.5  | 73.1  | (30) |
|                                                                        | 5%          | WB          | 64.7  | 64.5  | 68.0  | 71.4  | 76.3  | 81.1  | 83.9  | 82.8  | 79.3  | 75.3  | 69.4  | 67.4  | (31) |
|                                                                        |             | MCDB        | 68.0  | 70.4  | 74.7  | 78.1  | 82.8  | 86.0  | 90.5  | 87.0  | 83.5  | 79.3  | 72.3  | 70.9  | (32) |
|                                                                        | 10%         | WB          | 61.8  | 62.2  | 65.9  | 69.3  | 74.5  | 79.1  | 82.0  | 81.0  | 77.9  | 73.3  | 66.0  | 64.2  | (33) |
|                                                                        |             | MCDB        | 64.7  | 67.2  | 72.4  | 76.1  | 81.0  | 85.6  | 85.9  | 85.0  | 83.8  | 77.8  | 70.9  | 67.0  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 21.5  | 22.8  | 24.0  | 24.3  | 22.4  | 20.0  | 19.0  | 18.4  | 19.1  | 22.0  | 23.6  | 21.1  | (35) |
|                                                                        |             | MCDBR       | 24.3  | 26.8  | 27.2  | 27.0  | 25.2  | 23.0  | 21.7  | 20.8  | 22.1  | 23.2  | 25.8  | 22.9  | (36) |
|                                                                        |             | MCWBR       | 16.1  | 15.1  | 13.0  | 10.7  | 8.8   | 7.5   | 8.0   | 7.2   | 8.2   | 11.0  | 15.8  | 15.1  | (37) |
|                                                                        | 5% WB       | MCDBR       | 20.8  | 22.5  | 22.1  | 21.8  | 20.1  | 18.6  | 19.5  | 18.2  | 17.4  | 17.2  | 19.6  | 18.9  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.326 | 0.335 | 0.365 | 0.400 | 0.449 | 0.496 | 0.527 | 0.518 | 0.427 | 0.392 | 0.354 | 0.330 | (40) |
|                                                                        |             | taud        | 2.496 | 2.468 | 2.415 | 2.310 | 2.208 | 2.130 | 2.060 | 2.083 | 2.328 | 2.390 | 2.451 | 2.519 | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 284   | 292   | 290   | 283   | 268   | 254   | 246   | 246   | 266   | 268   | 270   | 276   | (42) |
|                                                                        |             | Edh at Noon | 28    | 31    | 36    | 41    | 46    | 49    | 53    | 51    | 38    | 34    | 29    | 26    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 931   | 1128  | 1469  | 1848  | 1988  | 1992  | 1943  | 1767  | 1518  | 1309  | 1025  | 813   | (44) |
|                                                                        | RadStd      | RadStd      | 47    | 108   | 120   | 123   | 127   | 149   | 129   | 98    | 126   | 144   | 74    | 72    | (45) |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m)  |      |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | N/A  | (46) |
| (47) | Regional (40 neighbors) | +1.04   | +1.33  | N/A     | N/A   | N/A   | N/A         | N/A   | -167  | +356  | +248 | (47) |

Nomenclature: See separate page